Problem A – The Tale of the Triangular Kingdom

Description

In the Triangular Kingdom, games were not played with cards or dice, but with black trikits – small, equilateral black triangular tiles, each etched with three numbers at its corners. Every black trikit was special. Its three numbers, always between 0 and 5, were carved in clockwise order, never decreasing as you went around the triangle. Some trikits repeated the same number more than once. Others were mixing different values – but no two tiles were ever exactly the same.

The elders of the kingdom had strict rules for how the game could be played:

- The first trikit could be placed freely on the table.
- Any trikit placed after that had to match one already on the table.
- A match happened only when two tiles shared a side with the same two numbers – corner to corner, side to side.

When a match occurred, the kingdom rejoiced, because points were earned. The score was simple: Every time two sides matched, the two matching numbers were added to the score. More matches meant more glory.

But there was a catch ... Not every tile could always be placed. Some trikits simply refused to fit anywhere, no matter how cleverly the table was arranged.

Given a collection of trikits – fewer than 20, to keep the challenge fair – you are appointed as the Master Strategist. Your task is not merely to play the game, but to play it perfectly:

- Choose which trikit to place first
- Decide which trikit to match, and in what order
- Accept that some trikits may remain unused

Your goal is to maximize the total score earned from all valid matches.

A trikit can only be played if it is the first one or if there is a matching trikit in the table, that is, both have a side with two matching numbers; see figure below.

In this exercise, the score in a trikit game is the sum of matching numbers. Given a set of trikits, the goal is to compute the maximum score that can be achieved. Note that it may not be possible to have all trikits matched.

Input

Each test case contains *n* lines, each of which describes a trikit as a sequence of three integer numbers in non-decreasing order.

Output

For each test case, print the maximum score that can be achieved.

Note: It would be wise to write a procedure that allows to print the current status of the game.

Constraints

- $n < 20$

Example

Example input:

```
0 3 4
0 0 2
2 4 4
0 1 2
2 3 4
1 4 5
0 2 2
2 2 4
```

Example output:

```
20
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